TEJAS NALADALA

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EDUCATION

University of Washington, Seattle Sep 2024 – Jun 2028

B.S. Electrical & Computer Engineering | Applied Mathematics (Double Major) | **GPA: 3.93/4.0**

Lavin Entrepreneurship Fellow - Selective 4-year founder development program (mentorship, venture capital exposure, leadership)

EXPERIENCE

PlasmaX - *Founder & CEO* May 2023 – Present

- Engineered the first industrial-scale continuous plasma-activated water disinfection system for post-harvest produce processing, replacing chlorine sanitation with a chemical-free process operating at $0.47/ton, delivering 2.5 $\log_{10}$ microbial reduction at 1.86 kW total system power
- Closed $2M safe seed round at $8M post-money valuation; sold first production unit to India's CSIR-CFTRI and contracted 10 revenue-generating system deployments across Indian commercial packhouses
- Filed patent on venturi-plasma integration architecture; co-authored 3 peer-reviewed publications in Journal of Physics D: Applied Physics and Innovative Food Science & Emerging Technologies
- Won UW Science & Technology Showcase Grand Prize ($2,500) and Best Pitch Award ($250), selected by 16 judges from Madrona Venture Group, Life Sciences Washington, and venture capital counsel
- Designed plasma reactor CAD, fabricated ionization chambers via CNC/VMC machining, laser/waterjet cutting, and injection molding; wired high-voltage control panels and coordinated procurement across Vietnamese manufacturing partners

NIIST (CSIR), India - *Research Intern* Jun 2024 – Mar 2025

Centre for Sustainable Energy Technologies - supervised by Dr. Suraj Soman (world-record DSC efficiency group)

- Fabricated and optimized dye-sensitized and perovskite solar cells, engineering ruthenium-TiO_2 photoanode interfaces, iodide/triiodide electrolyte charge dynamics, and polymer-based hole-transport layers over hybrid lead-halide perovskites
- Characterized 20+ device architectures using Dyenamo Toolbox, photo-induced absorption spectroscopy (PIA), IPCE, and XRD, validating photovoltaic performance metrics and identifying efficiency bottlenecks

SEAL Lab, University of Washington - *Research Associate* Mar – Nov 2025

Supervised by Prof. Alex Mamishev

- Designed embedded sensing architectures for 2 research grant proposals: a PPG-based drowsiness detection wearable for real-time physiological monitoring and a non-intrusive breakage-detection sensor system for U.S. Navy hull integrity assessment
- Authored grant sections, technical abstracts, and system specifications across 7-person interdisciplinary team; conducted literature analysis of 40+ papers on photoplethysmography signal processing and embedded anomaly detection

Cerulean Robotics - *Founder* Nov 2025 – Present

- Architecting open-source underwater ROV platform ($2–5K) with modular sensor payload system (water quality, sonar, thermal, manipulator modules), targeting 80-100m depth rating as accessible alternative to $10–50K commercial systems in the $3.5B AUV/ROV market
- Designed full system architecture: vectored 5-thruster configuration (Pixhawk/ArduSub autopilot), Raspberry Pi 5 HD vision pipeline, custom MOSFET-based power distribution for 6S Li-ion battery management, waterproofed acrylic pressure housings, and CAT6 tethered telemetry
- Secured $2,000 Buerk Center prototype funding; driving all mechanical CAD, embedded firmware, power electronics, waterproofing, business operations, and go-to-market strategy as sole technical founder.
- Engaged King County government officials and secured letter of intent for contracted underwater environmental auditing of Lake Sammamish, validating commercial demand for low-cost autonomous water-quality monitoring

Atticus AI - *Founder & Technical Lead* Jan 2026 – Present

- Architected document intelligence system with proprietary 4-stage PDF extraction pipeline and multi-model AI orchestration, automating coverage gap detection and anomaly identification across complex commercial insurance documents
- Built production Next.js 14 application with real-time multi-model inference pipeline, structured scoring engine, and automated report generation, processing full document analysis end-to-end in under 60 seconds
- Designed inference infrastructure with parallel multi-model API orchestration, token-aware context windowing across 100+ page documents, structured output validation via Zod schemas, and Redis caching layer reducing redundant LLM calls by 40% on repeat document patterns

PROJECTS

Forge - *Open-Source AI Agent Runtime* Feb 2026

- Built provider-agnostic AI agent orchestration framework (4,400+ LOC, Python/TypeScript) enabling declarative YAML agent definitions with automatic routing across 8 LLM providers (OpenAI, Anthropic, Google, Ollama, DeepSeek, Groq, Together AI, vLLM)
- Implemented modular architecture: agent runtime, LiteLLM model router, pluggable memory (SQLite + ChromaDB), tool registry with 6 built-in tools, and multi-agent orchestration (sequential, parallel, supervisor); FastAPI + WebSocket observability with Next.js dashboard; 150+ unique clones in first week

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PUBLICATIONS

3 peer-reviewed papers (2024) in J. Phys. D: Appl. Phys. and Innov. Food Sci. Emerg. Technol. + 1 international conference presentation (26th Food Innovation Asia, Bangkok). DOIs: 10.1088/1361-6463/ad77de | 10.1016/j.ifset.2024.103845 | 10.1016/j.ifset.2024.103840

TECHNICAL SKILLS

Embedded & Hardware: Circuit design (analog/digital), PCB design (KiCad), STM32/Arduino/Raspberry Pi, Pixhawk/ArduSub, MOSFET power electronics, sensor integration, oscilloscope/logic analyzer

AI & Machine Learning: LLM orchestration (OpenAI, Anthropic, Google), multi-agent systems, RAG pipelines, TensorFlow, OpenCV, ChromaDB, SQLite, LiteLLM, PDF/document AI, signal processing

Languages & Frameworks: Python, JavaScript/TypeScript, Java, C/C++, SQL, HTML/CSS, MATLAB/Simulink, Bash, Next.js, React, Node.js, FastAPI, REST APIs, WebSocket

Design & Fabrication: Fusion360, SolidWorks, Rhino 8, 3D printing, CNC/lathe, laser/waterjet cutting, soldering, injection molding

DevOps & Tools: Git/GitHub, Docker, Linux, ROS/Gazebo, LTspice, Jupyter

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LEADERSHIP & VENTURES

- Lavin Entrepreneurship Fellow - ~10% acceptance rate; 4-year intensive founder development with VC exposure, pitch training, and venture mentorship at University of Washington
- 3× Competition Winner - UW 2026 S&T Showcase Grand Prize + Best Pitch; Buerk Center Prototype Funding recipient